

Use the formula for the sum of the first n integers and/or the formula for the sum of a geometric sequence to evaluate the sums in 20-29 or to write them in closed form.

5.2.21. $5 + 10 + 15 + 20 + \dots + 300$

$$P(n) : \sum_{i=1}^n 5i = \frac{5n(n+1)}{2} \text{ for every integer } n \geq 1$$

$$5 + 10 + 15 + 20 + \dots + 300 = \sum_{i=1}^{60} 5i$$

$$5 + 10 + 15 + 20 + \dots + 300 = \frac{5n(n+1)}{2}$$

$$n = 60$$

$$\sum_{i=1}^{60} 5i = 5(1) + 5(2) + 5(3) + \dots + 5(60) = 9150$$

$$\frac{5n(n+1)}{2} = \frac{5(60)(61)}{2} = 9150$$

Therefore $P(60)$ is true.

Solution: $\sum_{i=1}^{60} 5i = \frac{5(60)(61)}{2} = 9150$